Charge, field & conductors

Start with the conductor: its boundary conditions determine the field.

Charge and superposition

Charge is conserved and quantised: q = ne. Add electric fields as vectors, but potentials as signed scalars.

$$\vec{E} = \sum_i \frac{q_i(\vec{r} - \vec{r}_i)}{4\pi\varepsilon_0 |\vec{r} - \vec{r}_i|^3}$$

$$V = \sum_i \frac{q_i}{4\pi\varepsilon_0 |\vec{r} - \vec{r}_i|}$$

WHY / Coulomb's inverse-square law applies directly to point charges. Extended conductors generally redistribute charge; do not replace them by centre charges unless symmetry or a distant-field approximation permits it.

Gauss and potential

$$\oint \vec{E} \cdot d\vec{A} = Q_{\text{enclosed}}/\varepsilon_0$$

$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l}$$

$$\vec{E} = -\nabla V; \quad \Phi = \frac{q\Omega}{4\pi\varepsilon_0}$$

The last result is flux through a patch subtending solid angle Omega at a point charge.

WHY / Gauss always holds; it yields E easily only when symmetry makes the field magnitude and direction known on a suitable surface.

Equilibrium conductor

Inside metal: E = 0; the entire connected conductor is equipotential. The external field is normal to its surface.

$$E_{\text{out, n}} = \sigma/\varepsilon_0$$

WHY / A tangential field would keep free charge moving. Equilibrium is reached only after charges rearrange to cancel it.

Excess charge resides on surfaces. Grounding fixes potential and allows charge exchange; isolation fixes net charge, not potential.

WATCH / A conductor need not be neutral when grounded, and need not have V = 0 when its internal field is zero.

Cavities and shielding

A charge +q in a cavity of a neutral isolated conductor induces -q on the cavity wall and +q on the outer surface.

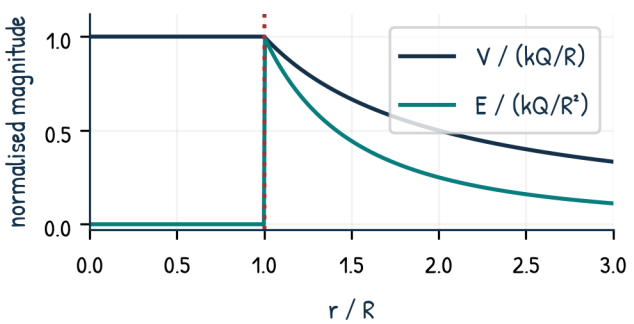
WHY / A Gaussian surface wholly inside the metal has zero flux. Its enclosed charge must therefore vanish.

An empty closed cavity is shielded from external static fields. With a charge inside, the cavity field is generally nonzero.

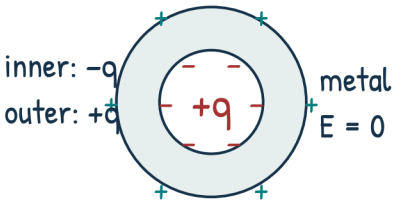
$$V_{\text{sphere}}(r < R) = \frac{Q}{4\pi\varepsilon_0 R}$$

$$V_{\text{sphere}}(r > R) = \frac{Q}{4\pi\varepsilon_0 r}$$

Conducting sphere: constant V, zero interior E



Neutral isolated shell with a cavity charge



REMEMBER / Zero field means constant potential, not necessarily zero potential.

Capacitance & parallel plates

Capacitance measures charge needed per volt, not how much charge is already present.

What C actually means

$$C = \frac{Q}{V_+ - V_-}; \quad 1\,\mathrm{F} = 1\,\mathrm{C/V}$$

For an ordinary two-terminal capacitor, plate charges are +Q and -Q. The pair's net charge is zero, not Q.

WHY / In a linear material, scaling every source charge scales every potential by the same factor. Their ratio is therefore set by geometry and permittivity.

An isolated conductor has its second terminal at infinity: C = Q/V, with V(infinity) = 0.

WATCH / Nonlinear dielectrics may require differential capacitance dQ/dV; constant C is then only an approximation.

Derive the plate formula

Two large facing plates of area A, gap d, filled with uniform permittivity epsilon:

$$E = \frac{Q}{\varepsilon A}, \quad V = Ed, \quad C = \frac{\varepsilon A}{d}$$

WHY / Fields of opposite plates add in the gap and cancel outside in the infinite-plate idealisation.

More overlap area stores more charge per volt. A larger gap needs more voltage for the same charge, so capacitance falls.

WATCH / Use overlap area, not the sum of both plate areas. Require d much smaller than the plate dimensions; edges produce fringing.

Metal inserted in the gap

A neutral floating metal slab of thickness t fills the overlap area without touching either plate.

$$C = \frac{\varepsilon_0 A}{d - t}$$

WHY / E = 0 inside the slab, so only the remaining vacuum thickness contributes to the voltage drop. Induced charges appear on its two faces.

Moving the slab changes the two vacuum gaps separately but not their sum: its position is irrelevant in the ideal model.

WATCH / Connecting the slab to a terminal changes the conductor topology. Do not reuse the floating-slab formula blindly.

Many plates and unequal charges

For n identical interleaved plates, alternate plates wired to the two terminals:

$$C = (n-1)\frac{\varepsilon A}{d}$$

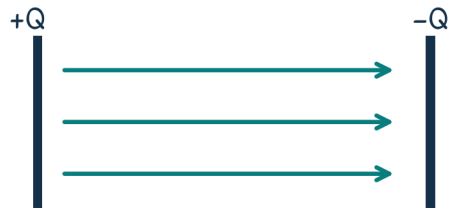
WHY / There are n - 1 active gaps, all with the same voltage, so their capacitances add in parallel.

If large equal plates carry net charges Q1 and Q2, facing charges are +(Q1-Q2)/2 and its negative, neglecting edges and external asymmetry.

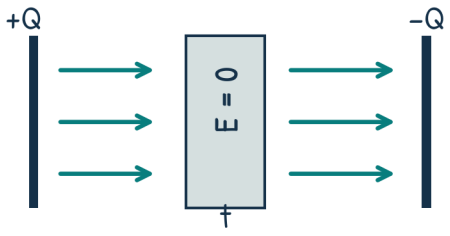
$$V_1 - V_2 = \frac{(Q_1 - Q_2)d}{2\varepsilon_0 A}$$

WATCH / Net plate charge can differ from charge on the facing surface.

Uniform gap field: E = V/d



Metal removes thickness t from the voltage drop



REMEMBER / Identify the two terminals and the potential reference before writing C = Q/V.

The geometry toolbox

One method everywhere: assign charge, find E, integrate the voltage, then divide.

Concentric spheres

Inner radius a, outer shell inner radius b; homogeneous dielectric epsilon between them.

$$E(r) = \frac{Q}{4\pi\varepsilon r^2}$$

$$V = \frac{Q}{4\pi\varepsilon}(1/a - 1/b)$$

$$C = 4\pi\varepsilon\frac{ab}{b-a}$$

WHY / Spherical Gaussian surfaces make E uniform on each radius. Only the field between a and b contributes to the terminal voltage.

As b tends to infinity: isolated sphere C = 4 pi epsilon a. For b-a much smaller than a, recover the plate formula with area 4 pi a squared.

Coaxial cylinders

Long coaxial conductors, radii a and b, length L; neglect end fields.

$$E(r) = \frac{Q}{2\pi\varepsilon L r}$$

$$C = \frac{2\pi\varepsilon L}{\ln(b/a)}$$

WHY / Cylindrical spreading makes E fall as 1/r. Integrating 1/r creates the logarithm.

Radial dielectric layers behave as series voltage drops:

$$\frac{1}{C/L} = \sum_i \frac{\ln(r_{i+1}/r_i)}{2\pi\varepsilon_i}$$

Two connected distant spheres

Spheres of radii a and b are joined by a long thin wire. Separation is much greater than either radius, so mutual effects are neglected.

$$V_a = V_b; \quad Q_a/a = Q_b/b$$

$$Q_a = Q\frac{a}{a+b}; \quad Q_b = Q\frac{b}{a+b}$$

$$\frac{E_a}{E_b} = \frac{b}{a}$$

WHY / A conducting connection enforces equal potentials, not equal charge. The smaller sphere has the larger surface field.

WATCH / Do not use these distant-sphere results for nearby spheres: induced charge makes the surface distributions nonuniform.

Useful composite geometries

Two distant spheres of radii a,b carrying opposite charges:

$$C \simeq 4\pi\varepsilon_0\frac{ab}{a+b}$$

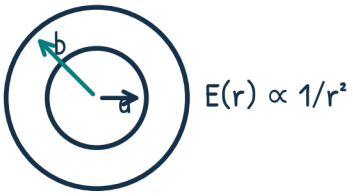
Only valid when separation greatly exceeds both radii. A conducting sphere a coated with dielectric K out to b:

$$\frac{1}{C} = \frac{(1/a - 1/b)/K + 1/b}{4\pi\varepsilon_0}$$

WHY / Add the voltage drop through the dielectric shell to the vacuum drop from b to infinity.

WATCH / Logarithms need dimensionless ratios. Isolated capacitance has scale epsilon0 times length; C per length has scale epsilon0.

Concentric spheres: cross-section



Coaxial cylinders: cross-section (length L)



REMEMBER / Derive standard geometries with Gauss and potential; connected conductors share potential, not charge.

Energy & the battery fork

The first question is not which formula to use, but what is being held fixed.

Why the factor one-half?

$$dU = V\,dq; \quad U = \int_0^Q \frac{q}{C}\,dq$$

$$U = \frac{Q^2}{2C} = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2$$

WHY / Voltage rises during charging. The average charging voltage is V/2, so the work is a triangle under the V-q graph, not a rectangle.

For a linear nondispersive dielectric:

$$u = \tfrac{1}{2}\varepsilon E^2; \quad U = \int u\,d\mathcal{V}$$

For a conductor, energy lies in the electric field outside the metal, not in its zero-field bulk.

Isolated: fix Q

After removing the battery and insulating each terminal, no charge can cross the terminal boundary.

$$U(x) = \frac{Q^2}{2C(x)}$$

$$F_x = -\left(\frac{dU}{dx}\right)_Q = \frac{Q^2 C'(x)}{2C(x)^2}$$

WHY / A mechanical change that increases C lowers field energy at fixed Q. The released energy can perform mechanical work.

Example: double a vacuum plate gap. C halves; Q and E stay fixed; V and U double. External work supplies the energy increase.

Connected: fix V

An ideal battery maintains terminal voltage by exchanging charge with the plates.

$$\Pi = U - VQ = -\tfrac{1}{2}C(x)V^2$$

$$F_x = -\left(\frac{d\Pi}{dx}\right)_V = \tfrac{1}{2}V^2 C'(x)$$

WHY / The battery gains or loses energy when charge flows. Differentiating field energy alone gives the wrong sign for the force.

WATCH / At the same instantaneous Q = CV, fixed-Q and fixed-V methods give the SAME force. The constraints affect how the state changes, not the force at that state.

Energy bookkeeping

For a slow fixed-V capacitance increase delta C:

$$W_{\text{battery}} = V^2\Delta C$$

$$\Delta U = \tfrac{1}{2}V^2\Delta C$$

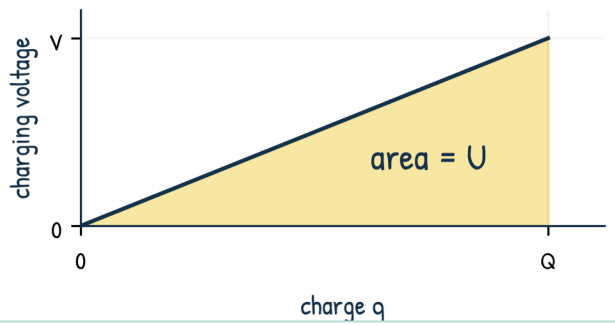
$$W_{\text{mech, out}} = \tfrac{1}{2}V^2\Delta C$$

WHY / Only half the battery work increases stored field energy; the rest can emerge as mechanical work in a quasistatic lossless motion.

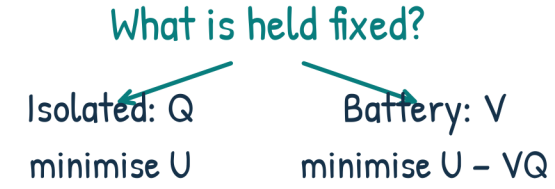
$$W_{\text{sources}} = \Delta U + H + W_{\text{mech, out}}$$

At fixed Q with no source or loss, mechanical work out equals -delta U.

Charging work is the area under V(q)



Choose the correct energy before taking a derivative



REMEMBER / At fixed V, the battery is part of the energy accounting—not an optional correction.

Pressure, insertion & liquid rise

Find C(x), state the constraint, then use virtual work instead of a difficult force integral.

The field acting on a plate

$$p = \frac{\sigma^2}{2\varepsilon_0} = \frac{1}{2}\varepsilon_0 E_{\text{out}}^2$$

This vacuum conductor pressure uses the field immediately outside the surface.

WHY / A surface element does not push itself. The field from other charges is half the total surface field, explaining the factor 1/2.

For ideal homogeneous parallel plates, attraction magnitude:

$$F = \frac{Q^2}{2\varepsilon A} = \frac{\varepsilon A V^2}{2d^2}$$

WATCH / Do not write F = QE using the total gap field; the correct magnitude is QE/2.

A partly inserted slab

Plate length L, width w, gap d; slab K fills the gap through inserted length x. Neglect fringe corrections.

$$C(x) = \frac{\varepsilon_0 w}{d}[L + (K-1)x]$$

$$F_{\text{in}} = \frac{\varepsilon_0 (K-1) w}{2d} V^2$$

WHY / The filled and empty areas are parallel capacitors. Increasing overlap increases C, so the force pulls the slab inward.

At fixed V, force is constant. At fixed Q, use instantaneous V = Q/C(x): the force decreases as the slab moves in.

Why a dielectric liquid climbs

Two vertical plates at fixed V draw dielectric liquid between them. Height h is above the outside reservoir; ignore capillarity.

$$h = \frac{\varepsilon_0 (K-1) V^2}{2\rho g d^2}$$

WHY / Raising the liquid replaces air with dielectric and increases C. Electric work pays for the gain in gravitational energy.

The electric force per unit plate width is epsilon0(K-1)V squared/(2d); balance it against the raised liquid's weight per width, rho g d h.

WATCH / Here h scales as V squared / d squared, not d to the minus fourth at fixed V.

Self-energy and mechanical checks

For an isolated conducting sphere of radius R:

$$U = \frac{Q^2}{8\pi\varepsilon_0 R}$$

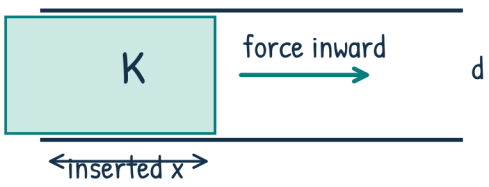
$$p = \frac{Q^2}{32\pi^2\varepsilon_0 R^4}$$

WHY / Expanding the sphere increases its capacitance and lowers its fixed-Q energy. The released energy corresponds to outward electric pressure.

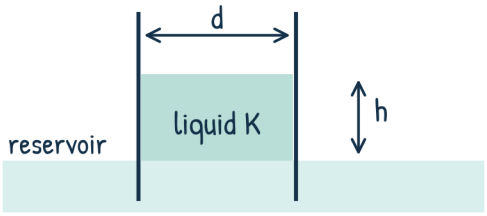
A uniformly charged solid sphere instead has U = 3Q squared/(20 pi epsilon0 R). Charge distribution matters.

WATCH / A symmetric charged sphere has no net translational force even though its surface experiences outward stress.

Insertion increases filled overlap area and C



Electric work lifts liquid against gravity



REMEMBER / A dielectric is pulled toward larger capacitance at fixed Q as well as fixed V.

Networks: charge on islands

Wires define conductor islands. Conservation applies to each isolated island, not to each capacitor.

Series: the real condition

$$\frac{1}{C_{\text{eq}}} = \sum_i \frac{1}{C_i}$$

$$V_i = Q/C_i; \quad V_1/V_2 = C_2/C_1$$

WHY / The intermediate conductor island cannot change its net charge. If initially neutral and attached only to two capacitors, their facing charges must balance.

WATCH / Equal series charge fails on a precharged island or an island with additional branches.

The smallest capacitor takes the largest voltage. Check every element's rating; total safe voltage is not always the sum of all individual ratings.

Parallel and energy division

$$C_{\text{eq}} = \sum_i C_i; \quad Q_i = C_i V$$

WHY / Capacitors connected between the same two nodes have identical voltage differences. Their terminal charges add.

Parallel: stored energy is proportional to C. In an initially neutral series chain, stored energy is inversely proportional to C.

$$\left(\frac{U_i}{U}\right)_{\text{series}} = \frac{C_{\text{eq}}}{C_i}$$

$$\left(\frac{U_i}{U}\right)_{\text{parallel}} = \frac{C_i}{C_{\text{eq}}}$$

WATCH / Series/parallel is about connections, not whether symbols are drawn in a straight line.

The method that never fails

Set terminal A to V and B to zero. Assign an unknown potential to each remaining floating island.

$$q_i^{\text{initial}} = \sum_j C_{ij}(V_i - V_j)$$

WHY / Each attached plate carries C times its voltage relative to the neighbouring node. Summing these signed plate charges gives the island's conserved net charge.

$$Q_A = \sum_j C_{Aj}(V_A - V_j)$$

For an initially uncharged passive network, Ceq = QA/V. Precharge can add an offset charge and must be tracked separately.

Sources and switching topology

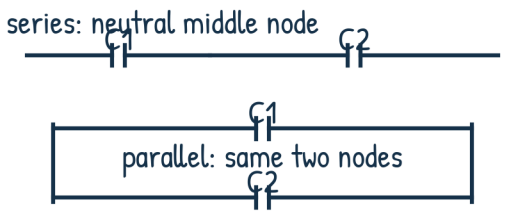
A grounded island has known V = 0; a floating island has known net charge. A battery imposes a difference of potentials.

WHY / A battery-connected island is not isolated: charge can enter through the cell. Do not impose both a prescribed potential and an unjustified neutrality condition.

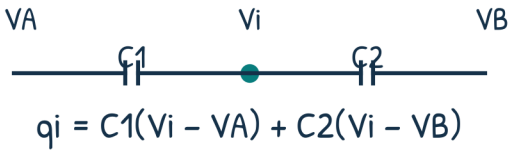
Before moving a switch: mark all plate charges and their signs. After moving it: redraw merged islands, add their initial charges, then impose final constraints.

WATCH / Conserve charge on the correct isolated conductor, not automatically the sum of the positive labels printed on capacitors.

Topology, not drawing orientation, defines the grouping



Add signed plate charges to conserve island charge



REMEMBER / Use the node-charge equation whenever a network cannot be reduced with confidence.

Bridges, symmetry & ladders

Simplifications are consequences of equal potentials—not guesses from a symmetric-looking drawing.

Balanced and unbalanced bridges

Upper path A-C1-X-C2-B; lower path A-C3-Y-C4-B; bridge C5 joins X to Y.

$$\frac{C_1}{C_2} = \frac{C_3}{C_4} \Rightarrow V_X = V_Y$$

WHY / Each path divides the terminal voltage in the same ratio. The bridge has zero voltage and carries no charge if initially uncharged.

Otherwise set VA = V, VB = 0 and solve:

$$C_1(V_X - V) + C_2 V_X + C_5(V_X - V_Y) = 0$$

$$C_3(V_Y - V) + C_4 V_Y + C_5(V_Y - V_X) = 0$$

Cube: fold only equal-potential nodes

All twelve cube edges have capacitance C. Across opposite vertices, the three neighbours of each terminal form equipotential groups.

$$3C,\ 6C,\ 3C \quad \text{inseries}$$

$$C_{\text{opposite}} = 6C/5$$

Other terminal choices: across an edge, 12C/7; across a face diagonal, 4C/3.

WHY / The excitation must preserve the symmetry used to merge nodes. Changing the terminal pair changes which nodes are equivalent.

WATCH / Never merge nodes just because their positions look similar on paper.

Delta-star for capacitors

Delta edges a=C12, b=C23, c=C31. Define S = ab + bc + ca. Equivalent star arms to nodes 1,2,3:

$$x = S/b, \quad y = S/c, \quad z = S/a$$

Reverse transformation, for star arms x,y,z:

$$C_{12} = \frac{xy}{x+y+z}$$

$$C_{23} = \frac{yz}{x+y+z}, \quad C_{31} = \frac{zx}{x+y+z}$$

WHY / Eliminate the neutral central node from its charge equation. These preserve the external charge-voltage relations.

WATCH / Do not copy resistor delta-star values unchanged.

Infinite self-similar ladder

One stage: series C, then shunt C to the lower rail, followed by the same infinite remainder X.

$$X = \frac{C(C+X)}{2C+X}$$

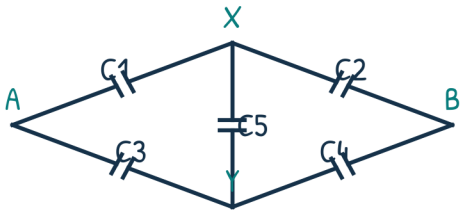
$$X^2 + CX - C^2 = 0$$

$$X = \frac{\sqrt{5}-1}{2}C$$

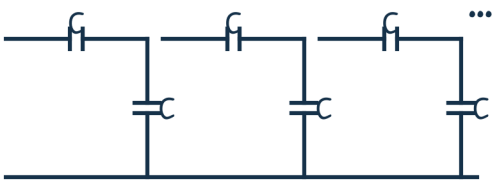
WHY / Removing one repeat unit leaves an identical infinite network. The negative quadratic root is rejected because passive equivalent capacitance is positive.

For finite ladders, start at the far end and repeatedly apply the same reduction. State which end is open or terminated.

Balance means VX = VY, so bridge voltage vanishes



Repeated capacitive ladder stages



REMEMBER / Always state the terminal choice: the same object can have different measured capacitances.

Dielectrics: field & geometry

A dielectric polarises; whether E changes depends on the constraint and the orientation.

Free versus bound charge

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}; \quad K = 1 + \chi_e$$

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon \vec{E}$$

$$\sigma_b = \vec{P} \cdot \hat{n}; \quad \rho_b = -\nabla \cdot \vec{P}$$

$$\oint \vec{D} \cdot d\vec{A} = Q_{\text{free}}$$

WHY / Molecular positive and negative charges shift slightly. Their bound surface charges oppose the original field at fixed free charge.

D is useful because its flux counts only free charge; it does not mean bound charge has no physical effect.

Interface conditions

$$D_{2n} - D_{1n} = \sigma_{\text{free}}$$

$$E_{2t} = E_{1t}$$

With no free interface charge, normal D is continuous and normal E varies inversely with permittivity.

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\varepsilon_1}{\varepsilon_2}$$

Angles are measured from the interface normal; assume no free surface charge for this refraction result.

WHY / Tangential E must match because an electrostatic closed-loop integral is zero. Normal D follows directly from a small Gaussian pillbox.

Layered gap: series

$$\frac{1}{C} = \sum_i \frac{t_i}{\varepsilon_i A}$$

$$D = Q_{\text{free}}/A; \quad E_i = D/\varepsilon_i$$

For vacuum plus a dielectric slab of thickness t:

$$C = \frac{\varepsilon_0 A}{d - t + t/K}$$

WHY / The same normal free-charge flux passes through every layer, while their voltage drops add. The weak-permittivity layer has the larger field.

Half the thickness filled: C = 2K C0/(K+1). As K tends to infinity, recover the metal-slab effective gap.

Area patches and full insertion

$$C = \sum_i \frac{\varepsilon_i A_i}{d}$$

Half the area filled through the full gap: C = C0(K+1)/2. Each patch sees the same voltage and E = V/d.

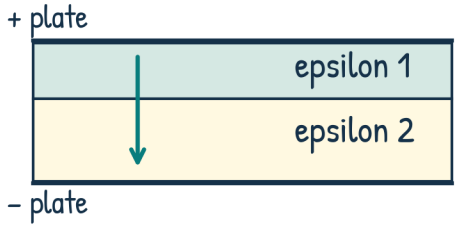
WHY / Here charges add at common voltage, unlike thickness layers where voltage drops add at common D.

Full K insertion, battery removed: Q fixed, C grows K-fold; V, E and U fall by K.

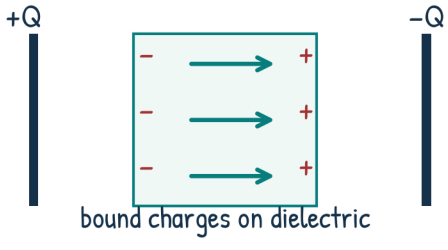
Full K insertion, battery connected: V and E fixed; C, Q and U grow K-fold.

WATCH / A dielectric does NOT always reduce E. At fixed V in a completely filled ideal gap, E remains V/d.

Series layers: same normal D; voltage drops add



Dipoles point with E; bound faces oppose it



REMEMBER / Layers along the field add voltage drops; side-by-side patches add terminal charges.

DC circuits & one RC transient

Every single-mode transient is determined by its starting value, final value and time constant.

The three-number method

$$V_C(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau}$$

$$\tau = R_{\text{th}}C$$

Find V0 from the pre-switch state, V infinity from the final DC resistor circuit, and Rth looking into the capacitor terminals.

WHY / Replacing the rest of a linear single-capacitor circuit by its Thevenin source leaves the same first-order differential equation.

WATCH / Voltage is continuous across switching only when there is no impulse current. Conflicting ideal source constraints need a more physical model.

Find the Thevenin resistance

Remove the capacitor. Find the open-circuit terminal voltage for V infinity. To find Rth, short independent voltage sources and open independent current sources.

Keep dependent sources active; apply a test source if necessary.

WHY / The resistance seen from the capacitor determines how quickly a small voltage error can drive restoring current.

$$V_\infty = V_s\frac{R_2}{R_1 + R_2}$$

$$R_{\text{th}} = R_1 \parallel R_2$$

These results apply to a capacitor across R2 in the divider drawn below.

Charging from an ideal cell

Initially uncharged C, source emf E, single series R:

$$V_C = E(1 - e^{-t/(RC)})$$

$$q = CE(1 - e^{-t/(RC)}), \quad i = \frac{E}{R}e^{-t/(RC)}$$

WHY / As capacitor voltage rises, the voltage left across the resistor falls, so the charging current decays.

At one time constant: voltage reaches 63.2% and current falls to 36.8%. After five time constants: 99.3% charged.

$$t = -RC\ln(1 - V_C/E)$$

Discharge and energy decay

$$V_C = V_0e^{-t/(RC)}; \quad |i| = \frac{V_0}{R}e^{-t/(RC)}$$

$$U = U_0e^{-2t/(RC)}$$

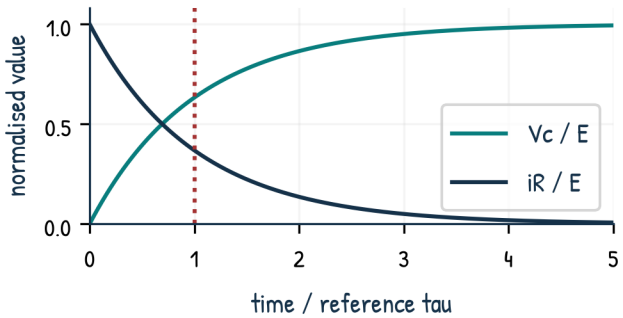
$$t_{1/2,\,V} = RC\ln 2; \quad t_{1/2,\,U} = \tfrac{1}{2}RC\ln 2$$

WHY / Energy is proportional to voltage squared, so it decays twice as fast in the exponential exponent.

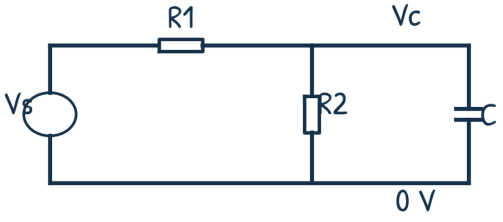
If leakage resistance Rp remains across a capacitor, it slowly discharges with time constant Rp C. At exact infinite time its voltage reaches zero unless a source maintains it.

WATCH / Time constant belongs to the actual post-switch topology, not automatically to the resistor nearest the capacitor.

Charge rises as current decays



C sees the divider voltage and R1 parallel R2



REMEMBER / A capacitor is open at DC steady state, but its final voltage can be nonzero.

Charge sharing, heat & branch charge

Conserve charge on isolated islands; use energy separately to find the heat released.

Joining two capacitors

Choose a common polarity convention; V1 and V2 are signed initial voltages. Join corresponding plates through resistance.

$$V_f = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$C_s = \frac{C_1C_2}{C_1 + C_2}$$

$$H = \tfrac{1}{2}C_s(V_1 - V_2)^2$$

WHY / The connected conductor islands conserve charge, but current dissipates energy while the voltage mismatch disappears.

Opposite-polarity connection is handled by a negative initial voltage, not a separate memorised formula.

Time scale and the paradox

$$V_1(t) - V_2(t) = (V_1 - V_2)e^{-t/(RC_s)}$$

The connecting resistance R sets the redistribution time, not the total energy lost after settling.

WHY / Smaller R gives a larger current for a shorter time. The integral of i squared R remains the same.

If several series resistors carry the same current, their heats divide in the resistance ratio.

WATCH / An ideal zero-resistance, zero-inductance connection is singular. Real leads radiate and have inductance; an ideal lossless LC circuit oscillates instead of settling.

Step charging energy balance

$$W_{\text{cell}} = CE^2; \quad U_f = H = \tfrac{1}{2}CE^2$$

For a single RC settling from V0 toward a constant Thevenin voltage Vth:

$$H = \tfrac{1}{2}C(V_0 - V_{\text{th}})^2$$

$$H(t) = \tfrac{1}{2}CE^2(1 - e^{-2t/(RC)})$$

The time-dependent expression assumes charging an initially uncharged C from E.

WHY / At Vc = E/2, time is RC ln2 and 3/4 of the total heat has already been released. Initial current is largest.

Charge through a cell or branch

Choose a conductor boundary crossed by only the branch whose transported charge is wanted. Add every capacitor-plate charge inside, with signs.

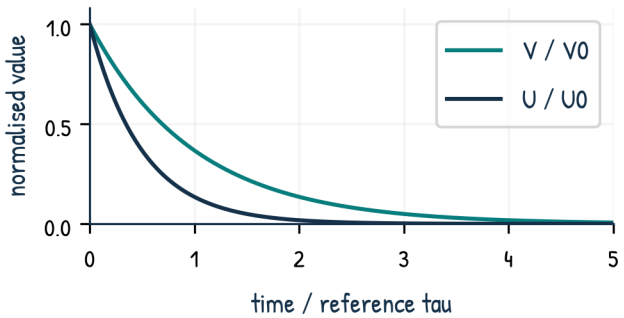
$$Q_{\text{branch, in}} = \Delta\left(\sum_{\text{inside}} q_{\text{plate}}\right)$$

WHY / The change of net enclosed charge must equal net charge entering through that branch. This gives total transported charge directly from endpoint states.

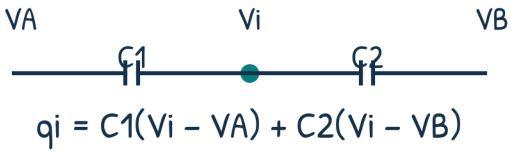
The battery work is its emf times the signed charge it supplies. For several sources, sum their separate contributions.

WATCH / Do not use Ceq times delta V blindly after rewiring: island boundaries and initial charges matter.

Energy decays faster than voltage



Add signed plate charges to conserve island charge



qi = C1(Vi − VA) + C2(Vi − VB)

REMEMBER / Charge conservation fixes the final voltage; it does not imply field-energy conservation.

JEE revision & safe-use checklist

Finish the capacitor chapter here: core concepts, useful applications and checks before moving to extensions.

The reliable solving sequence

1. Define terminals, voltage polarity and what Q means.
2. Redraw all wires as conductor islands; mark grounded, floating and battery-connected parts.
3. State what remains fixed: free charge, voltage, geometry or a source constraint.
4. Reduce only genuine series/parallel groups; otherwise write node-charge equations.
5. For mechanics, find C(x) and differentiate the appropriate energy. For a transient, find V0, V infinity and Rth.

WHY / Most difficult-looking capacitor questions are ordinary conservation laws applied to a carefully chosen system boundary.

Dielectric strength and ratings

$$V_{\max} \simeq E_b d; \quad u_{\max} = \tfrac{1}{2}\varepsilon E_b^2$$

Use a roughly uniform field and a linear dielectric. Choose a working voltage below the breakdown rating; avoid treating the maximum as a safe operating target.

WHY / Breakdown depends on local field. A sharp edge or defect can fail even when the average V/d appears acceptable.

In series, the smallest capacitor takes the largest voltage for equal charge. Check each capacitor separately, not only the total applied voltage.

WATCH / A larger dielectric constant increases C, but high energy storage also requires high dielectric strength.

Six quick sanity checks

Dimensions: C has farads; an isolated body's C scales as epsilon0 times length.

Limits: K to 1 should recover vacuum; outer spherical radius to infinity should recover an isolated sphere.

Signs: dissipated heat cannot be negative; force on an inserted dielectric points toward larger C.

Endpoints: a transient must begin and end at the values implied by the actual circuit.

Stability: effective energy curvature must be positive.

Ratings: test local fields and each element's voltage, not only total applied voltage.

Numbers and conceptual anchors

$$\varepsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F/m}$$

$$4\pi\varepsilon_0 = 1.113 \times 10^{-10}\ \mathrm{F/m}$$

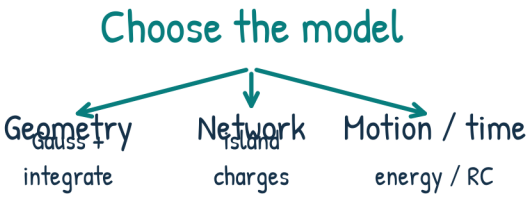
A 1 cm isolated sphere has C about 1.11 pF. Air breakdown is roughly 3 kV/mm for macroscopic gaps, but condition-dependent.

Fixed Q, full dielectric insertion: voltage and field energy decrease. Fixed V: stored energy increases because the source supplies charge.

WHY / Same state Q = CV means same instantaneous force, even though the two constraints lead to different subsequent motion.

WATCH / These sheets complement the full practice paper.

Pick a physical method before doing algebra



Charge sets the final voltage; energy sets the heat

Quick self-test

2 µF at 100 V + 3 µF at 0 V

Same polarity: Vf = 40 V

Heat = 6 mJ. Can you derive both?

REMEMBER / This is the end of Part I. The remaining sheets are optional extensions, not prerequisites for the JEE core.

Microscopic & dispersive dielectrics

EXTENSION: microscopic material models and frequency-dependent response, beyond the core capacitor chapter.

From dipoles to permittivity

$$p = \alpha E_{\text{local}}; \quad E_{\text{local}} = E + P/(3\varepsilon_0)$$

$$\frac{K-1}{K+2} = \frac{N\alpha}{3\varepsilon_0}$$

This Lorentz / Clausius-Mossotti model assumes an isotropic local-field description.

$$\alpha_e = \frac{e^2}{m\omega_0^2}, \quad \alpha_{\text{orient}} = \frac{p_0^2}{3k_BT}$$

WHY / Electronic polarisation is charge-cloud distortion; orientational polarisation is partial alignment of permanent dipoles against thermal agitation.

Orientational response weakens as temperature rises at fixed number density.

Debye relaxation and losses

Use time dependence exp(i omega t) and K* = K' - i K double-prime.

$$K^* = K_\infty + \frac{K_s - K_\infty}{1 + i\omega\tau}$$

$$K'' = \frac{(K_s - K_\infty)\omega\tau}{1 + (\omega\tau)^2}$$

$$\tan\delta = K''/K'$$

WHY / Dipoles cannot follow a rapidly changing field instantly. Their lag dissipates energy and makes the apparent permittivity frequency-dependent.

K double-prime peaks at omega tau = 1; tan delta peaks instead at sqrt(Ks/Kinfinity).

Heat, leakage and memory

$$P_{\text{loss}}/\mathcal{V} = \omega\varepsilon_0 K' \tan\delta\, E_{\text{rms}}^2$$

$$\tan\delta \simeq \omega C R_s \simeq \frac{1}{\omega C R_p}$$

Small-loss series ESR and parallel leakage are approximate equivalent models at a chosen frequency.

WHY / A real capacitor stores energy and dissipates some. Dielectric absorption adds slow internal modes: a discharged capacitor can recover voltage.

Ferroelectrics show hysteresis and bias/temperature dependence. Piezoelectric strain S = d33 E; free thickness change is d33 V.

WATCH / Stored energy is not generally CV squared/2 when C is strongly nonlinear.

Gas breakdown and small gaps

$$V_b = f(pd) \quad \text{(Paschen law)}$$

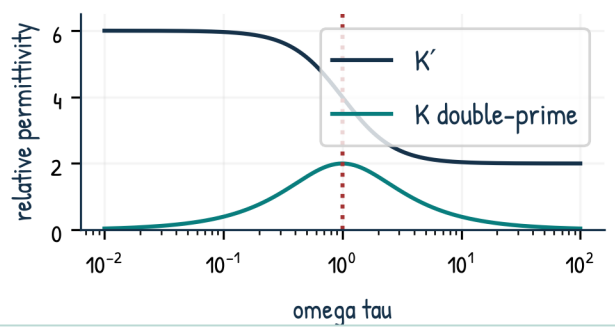
For classical gas discharge, breakdown depends on the product of gas pressure p and gap d. Air has a minimum near 330 V under the relevant discharge assumptions.

WHY / Avalanche growth competes with the probability of electron collisions. The pressure-gap product controls the number of opportunities for ionisation.

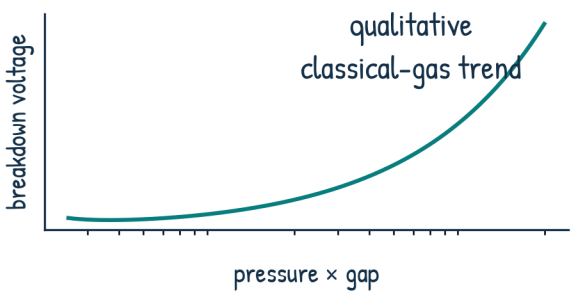
At very small gaps, field emission and tunnelling can invalidate a classical-gas model. Sharp features and voids concentrate local fields.

WATCH / The Paschen minimum is not an absolute voltage limit for every microscopic discharge mechanism.

Loss lags the response; K double-prime peaks at 1



Paschen minimum is not a universal microgap rule



REMEMBER / Distinguish intrinsic electrostatic storage from leakage, delayed polarisation and breakdown.

AC, dividers & timing circuits

EXTENSION: filters, compensated probes, threshold oscillators and slow-charging applications.

Impedance and the low-pass

$$i = C\frac{dV_C}{dt}; \quad Z_C = \frac{1}{i\omega C}$$

Sinusoidal current leads capacitor voltage by 90 degrees. For a series RC, output across C:

$$H(\omega) = \frac{1}{1 + i\omega RC}$$

$$|H| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$f_c = \frac{1}{2\pi RC}; \quad \phi = -\arctan(\omega RC)$$

WHY / At high frequency, C has small impedance and its output voltage is small. At cutoff: -3 dB and -45 degrees.

High-pass and compensated divider

Output across R in the same series RC:

$$H = \frac{i\omega RC}{1 + i\omega RC}$$

Compensated divider: upper arm R1 parallel C1, lower arm R2 parallel C2; output across lower arm.

$$\frac{V(0^+)}{V_s} = \frac{C_1}{C_1 + C_2}, \quad \frac{V(\infty)}{V_s} = \frac{R_2}{R_1 + R_2}$$

$$R_1C_1 = R_2C_2$$

WHY / Matching these time constants makes the fast capacitive division equal the slow resistive division, eliminating step-response distortion.

Relaxation timing

Charge from Vlo to Vhi toward Vs through R:

$$t_c = RC\ln\frac{V_s - V_{\text{lo}}}{V_s - V_{\text{hi}}}$$

Discharge from Vhi to Vlo toward zero through Rd:

$$t_d = R_dC\ln(V_{\text{hi}}/V_{\text{lo}})$$

For thresholds Vs/3 and 2Vs/3 and equal charge/discharge resistances: T = 2RC ln2.

WHY / A switching threshold converts a monotonic exponential into a repeated oscillation. The asymptotic voltage must lie beyond the threshold to reach it.

Ripple, gentle charging and noise

$$\Delta V \simeq I_{\text{load}}\Delta t/C$$

A reservoir capacitor supplies load charge between recharge events; choose C from the allowed ripple and longest recharge interval.

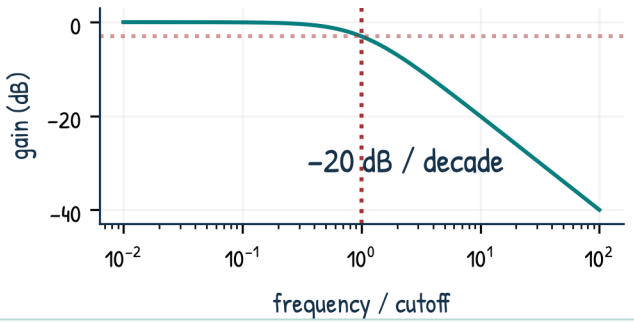
$$H_{\text{ramp}} \simeq RC^2V^2/T \quad (T \gg RC)$$

WHY / Slowly ramping the source keeps resistor current small, avoiding the unavoidable half-energy loss of a sudden voltage step.

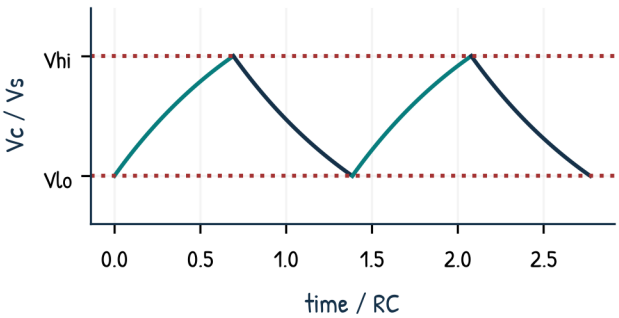
$$V_{\text{rms, noise}} = \sqrt{k_BT/C}$$

At 300 K and 1 pF: about 64 microvolt. Above self-resonance, lead inductance can dominate: a real capacitor is no longer capacitive.

RC low-pass across C: one pole, one time constant



Threshold switching repeats charge and discharge



REMEMBER / Always identify where the output voltage is measured before choosing a filter formula.

Coupled RC & specialised models

EXTENSION: coupled modes, measurement loading and exact wire-to-wire / wire-to-plane capacitances.

Several independent storage modes

$$\mathbf{C}\dot{\mathbf{V}} + \mathbf{G}\mathbf{V} = \mathbf{b}$$

Use independent node voltages after applying source and charge constraints. The matrices describe capacitive storage and resistive conduction.

$$V(t) = V_\infty + \sum_k A_k e^{-\lambda_k t}$$

WHY / Each independent relaxation mode has its own rate. A measured voltage can contain several exponentials, or hide a mode through symmetry.

For an invertible reduced capacitance matrix, rates are eigenvalues of C inverse G. Counting drawn capacitors alone is not sufficient.

An equal-R, equal-C example

Two nodes each have C to ground. An ideal source drives node 1 through R; another R joins nodes 1 and 2 (diagram below).

$$RC\dot{V}_1 = V_s - 2V_1 + V_2$$

$$RC\dot{V}_2 = V_1 - V_2$$

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2RC}$$

WHY / The two node voltages can relax in different combinations. The eigenvalues of the conductance-storage system give two distinct time scales.

After a constant source step, both final voltages equal Vs. Initial voltages determine how strongly each mode appears.

Measuring can change the voltage

An isolated capacitor Cs at voltage V0 is connected to an initially uncharged meter input capacitance Cm. Neglect leakage during the reading.

$$V_{\text{read}} = V_0 \frac{C_s}{C_s + C_m}$$

WHY / The meter needs charge to raise its own input voltage. Charge sharing lowers the original source-capacitor voltage.

$$V_{\text{rms, noise}} = \sqrt{k_B T / C}$$

The thermal voltage scale is about 64 microvolt at 300 K and 1 pF. Small capacitance is sensitive, but more vulnerable to loading and noise.

Wires and the ground plane

Two equal long wires: radius a, centre spacing D > 2a, homogeneous surrounding dielectric.

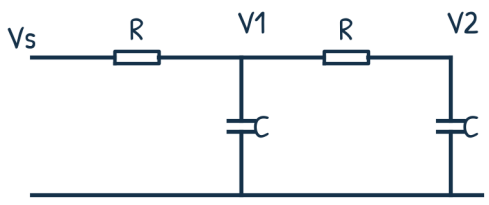
$$\frac{C}{L} = \frac{\pi\varepsilon}{\operatorname{arcosh}(D/2a)}$$

For D much larger than a, denominator becomes ln(D/a). One wire at centre height h > a above a grounded plane:

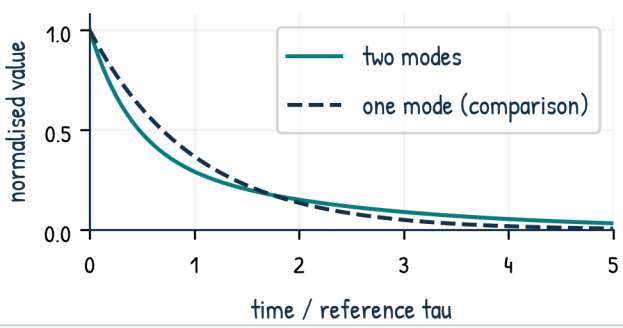
$$\frac{C}{L} = \frac{2\pi\varepsilon}{\operatorname{arcosh}(h/a)}$$

WHY / The ground plane is an equipotential midway between a wire and its opposite image. The wire-to-plane voltage is half the pair voltage.

Two storage nodes give two relaxation rates



Several storage modes need not share one time constant



REMEMBER / A single exponential is a model choice: identify the independent dynamic variables before assigning one time constant.

Matrices, reciprocity & images

Advanced tools simplify boundary conditions; they do not replace charge and energy bookkeeping.

Many conductors and reciprocity

$$Q_i = \sum_j C_{ij} V_j; \quad C_{ij} = C_{ji}$$

$$U = \tfrac{1}{2}\sum_i Q_i V_i = \tfrac{1}{2}\mathbf{V}^T\mathbf{C}\mathbf{V}$$

With a fixed reference: diagonal terms positive, off-diagonal terms nonpositive. If invertible, P = C inverse.

$$\sum_i Q_i V'_i = \sum_i Q'_i V_i$$

WHY / Reciprocity follows from symmetric C for the same geometry and linear dielectric.

Floating fixes Q; grounding fixes V.

Weighting potential and signal

Set the selected electrode to unit potential and every other electrode to zero, with the moving charge removed. Call the resulting potential phi_w.

$$\Delta Q_{\text{electrode}} = -q\Delta\phi_w$$

$$i_{\text{electrode}} = -q\vec{v}\cdot\nabla\phi_w$$

WHY / The weighting field measures how strongly each point couples to the selected electrode. It is not the actual driving field.

For parallel plates separated by d, phi_w = x/d when the selected electrode is at x = d. Current sign here means the rate of induced electrode charge.

Charge above a grounded plane

Replace a grounded plane by image -q at the mirror point. A real +q at height h feels:

$$|F| = \frac{q^2}{16\pi\varepsilon_0 h^2}$$

$$U = -\frac{q^2}{16\pi\varepsilon_0 h}$$

$$\sigma(\rho) = -\frac{qh}{2\pi(\rho^2 + h^2)^{3/2}}$$

WHY / The opposite image enforces V = 0 on the plane. The real force equals the image force, but induction energy is one-half q times image potential.

WATCH / Use separation 2h in Coulomb's force; the field is not that of an image only h away.

Grounded versus neutral sphere

Real q at distance d > a from a grounded sphere's centre. Place an image on the same radial line:

$$q' = -qa/d; \quad r' = a^2/d$$

$$|F| = \frac{q^2 ad}{4\pi\varepsilon_0(d^2 - a^2)^2}$$

WHY / The chosen image makes the entire spherical surface an equipotential at zero, not merely its nearest point.

For an isolated neutral sphere, add a central image +qa/d so total image charge is zero. Its force contribution must also be included.

WATCH / A grounded-sphere force formula is not automatically valid for a neutral floating sphere.

Real-image separation is 2h

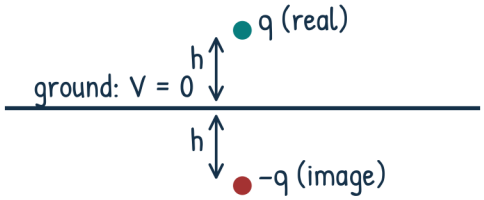
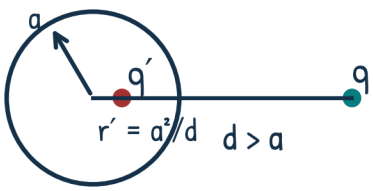


Image lies inside sphere; real charge is outside



REMEMBER / Image charges are mathematical replacements in the physical region—not extra real charges.

Shape, conformal maps & pores

Geometry controls both charge storage and where the electric field becomes dangerously large.

Spheroids: needle to disc

Semiaxis c along the symmetry axis, transverse semiaxis a. Prolate c > a; let f squared = c squared − a squared.

$$C = \frac{4\pi\varepsilon_0 f}{\ln[(c+f)/a]}$$

Oblate a > c: f squared = a squared − c squared; eta = arctan(f/c).

$$C = 4\pi\varepsilon_0 f/\eta$$

Limits: sphere C = 4 pi epsilon0 a; thin disc C = 8 epsilon0 a; long needle C approximately 4 pi epsilon0 c / ln(2c/a).

WHY / Tip curvature concentrates charge and local field. For an ellipsoidal inclusion the depolarising factor similarly controls its internal field.

Ellipsoidal inclusions and voids

$$E_{\text{in}} = \frac{E_0}{1 + L(\varepsilon_i/\varepsilon_m - 1)}$$

$$L_1 + L_2 + L_3 = 1$$

Here L is the depolarising factor along the applied field. Sphere: 1/3; axial needle: approaches 0; thin disc normal to field: approaches 1.

WHY / Polarisation charges on an inclusion's surface create an opposing field whose strength depends on shape.

An air disc-shaped crack normal to E in a K host sees about K E0. An axial needle-shaped void sees about E0. Thin cracks can therefore be much worse defects.

Wedges and corner fields

Two conducting faces at angles 0 and beta, radii r0 to R, held at 0 and V; beta in radians.

$$\phi = V\theta/\beta; \quad E_\theta = -V/(\beta r)$$

$$C' = \frac{\varepsilon}{\beta}\ln(R/r_0)$$

$$E_{\text{corner}} \propto r^{\,\pi/\beta - 1}$$

In the corner expression, beta is the vacuum opening angle between equipotential conductor faces. A knife edge has beta = 2 pi and E proportional to r to the minus one-half.

WHY / An analytic conformal map preserves 2D Laplace solutions; changing coordinates can turn a hard conductor shape into a simple strip.

Porous electrodes: delayed storage

Pore resistance per length r, capacitance per length c:

$$c\frac{\partial V}{\partial t} = \frac{1}{r}\frac{\partial^2 V}{\partial x^2}$$

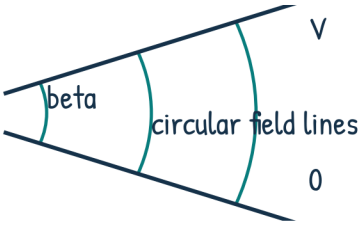
$$\ell_{\text{reached}} \sim \sqrt{t/(rc)}$$

$$Z_\infty = (1-i)\sqrt{r/(2\omega c)}$$

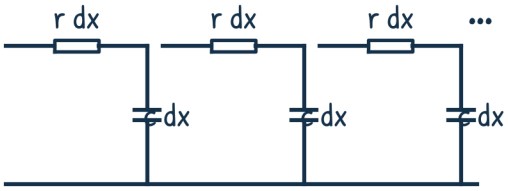
WHY / Charge must travel through resistance to reach deeper internal surface. Only part of the large geometric capacitance is accessible at short times.

Initially Q scales as square root of t. Full charging needs t of order rc L squared. A finite closed pore has low-frequency C = cL and ESR = rL/3.

Wedge: E falls as 1/r away from the apex



Distributed resistance delays access to deep pore walls



REMEMBER / A large total capacitance does not imply a small local field or rapid access to all stored charge.

Pull-in, oscillations & charged drops

EXTENSION: electromechanical instability and charged-drop shape stability, not a prerequisite for JEE capacitor problems.

Voltage-driven movable plate

Spring stiffness k, initial gap d0, closing displacement x; plate area A and homogeneous epsilon.

$$\Pi = \tfrac{1}{2}kx^2 - \frac{\varepsilon A V^2}{2(d_0 - x)}$$

$$kx = \frac{\varepsilon A V^2}{2(d_0 - x)^2}$$

WHY / Closing the gap strengthens voltage-driven attraction faster than the spring's restoring force. The stable and unstable equilibria can merge.

$$\Pi'' = k - \frac{\varepsilon A V^2}{(d_0 - x)^3} > 0$$

Pull-in and small oscillations

$$x_{\mathrm{pi}} = d_0/3$$

$$V_{\mathrm{pi}}^2 = \frac{8kd_0^3}{27\varepsilon A}$$

$$\omega^2 = \frac{1}{m}\left[k - \frac{\varepsilon A V^2}{(d_0 - x_{\mathrm{eq}})^3}\right]$$

WHY / Near pull-in, effective spring stiffness tends to zero, so the small-oscillation frequency softens to zero.

Use this formula only about a stable equilibrium and for small displacement. The one-third stroke limit assumes ideal parallel plates and a linear spring.

What changes at fixed charge?

$$U_Q = \frac{Q^2(d_0 - x)}{2\varepsilon A}$$

$$F = \frac{Q^2}{2\varepsilon A}; \quad x_{\mathrm{eq}} = \frac{Q^2}{2k\varepsilon A}$$

WHY / The ideal fixed-Q electric force is independent of gap. It shifts the spring equilibrium but does not soften its stiffness.

$$\omega = \sqrt{k/m}$$

There is no voltage-type fold before contact in this ideal model. Contact, fringing or charge leakage can invalidate it.

WATCH / Do not carry the voltage-driven pull-in result into a charge-controlled problem.

Rayleigh limit of a liquid drop

A conducting drop of radius R and surface tension gamma in vacuum has electric surface pressure:

$$p_e = \frac{Q^2}{32\pi^2\varepsilon_0 R^4}$$

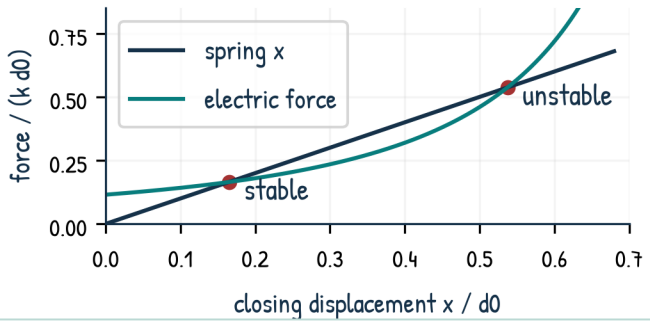
$$Q_R^2 = 64\pi^2\varepsilon_0\gamma R^3$$

$$V_R = 2\sqrt{\gamma R/\varepsilon_0}$$

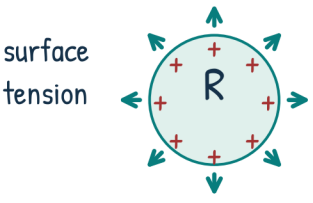
WHY / Above the Rayleigh limit, a shape deformation can lower total energy: surface tension can no longer stabilise the charged sphere.

This is a shape-instability result, not simply radial expansion at fixed volume. Real drops may also be limited by gas breakdown or charge emission.

Fixed V: lower-x crossing stable; higher-x unstable



Electric stress destabilises; surface tension stabilises



REMEMBER / A force-balance solution is not enough: test stability under the actual electrical constraint.